Ex. No. 4	SET OPERATIONS
Date of Exercise	19/08/2026

Aim

To execute SET operators (UNION, INTERSECT, MINUS) and the IN operator to combine and compare results from multiple queries.

Description

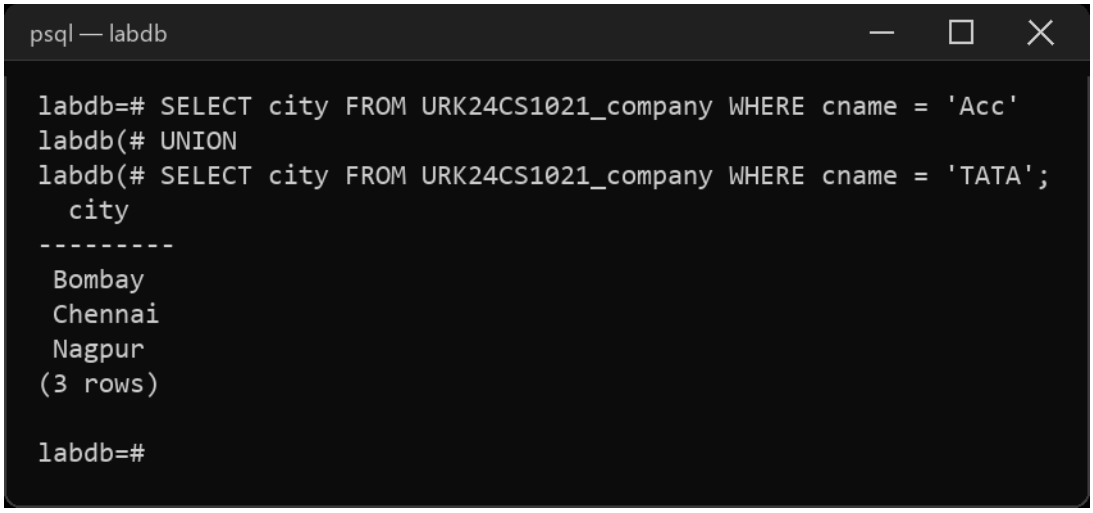
Set operators combine the results of two or more SELECT queries that return the same number and type of columns. UNION returns the distinct rows from either query, while UNION ALL keeps duplicates. INTERSECT returns only the rows common to both queries, and MINUS (EXCEPT in PostgreSQL) returns the rows in the first query that are absent from the second. The IN operator tests whether a value matches any value returned by a subquery, and is often used as an alternative to a join.

1. Display the cities of the companies Acc and TATA.

Sample Code:

```
SELECT city FROM URK24CS1021_company WHERE cname = 'Acc'
UNION
SELECT city FROM URK24CS1021_company WHERE cname = 'TATA';
```

Sample Output:



2. Display the names of the city that is common to companies Acc and TATA.

Sample Code:

```
SELECT city FROM URK24CS1021_company WHERE cname = 'Acc'
INTERSECT
SELECT city FROM URK24CS1021_company WHERE cname = 'TATA';
```

Sample Output:

```
psql — labdb

labdb=# SELECT city FROM URK24CS1021_company WHERE cname = 'Acc'
labdb(# INTERSECT
labdb(# SELECT city FROM URK24CS1021_company WHERE cname = 'TATA';
  city
---------
 Chennai
(1 row)

labdb=#
```

3. Find the cities where Acc is located, but not TATA.

Sample Code:

```
SELECT city FROM URK24CS1021_company WHERE cname = 'Acc'
EXCEPT
SELECT city FROM URK24CS1021_company WHERE cname = 'TATA';
```

Sample Output:

```
psql — labdb

labdb=# SELECT city FROM URK24CS1021_company WHERE cname = 'Acc'
labdb(# EXCEPT
labdb(# SELECT city FROM URK24CS1021_company WHERE cname = 'TATA';
  city
--------
 Nagpur
(1 row)

labdb=#
```

4. Display the names of employees living in Nagpur, and working with company Acc.

Sample Code:

```
SELECT ename FROM URK24CS1021_employee WHERE city = 'Nagpur'
INTERSECT
SELECT ename FROM URK24CS1021_emp_company WHERE cname = 'Acc';
```

Sample Output:

```
psql — labdb

labdb=# SELECT ename FROM URK24CS1021_employee WHERE city = 'Nagpur'
labdb(# INTERSECT
labdb(# SELECT ename FROM URK24CS1021_emp_company WHERE cname = 'Acc';
 ename
-------
 Anil
(1 row)

labdb=#
```

5. Display the names of employees living in Nagpur, but not working in company Acc.

Sample Code:

```
SELECT ename FROM URK24CS1021_employee WHERE city = 'Nagpur'
EXCEPT
SELECT ename FROM URK24CS1021_emp_company WHERE cname = 'Acc';
```

Sample Output:

```
psql — labdb

labdb=# SELECT ename FROM URK24CS1021_employee WHERE city = 'Nagpur'
labdb(# EXCEPT
labdb(# SELECT ename FROM URK24CS1021_emp_company WHERE cname = 'Acc';
 ename
-------
 Ajay
(1 row)

labdb=#
```

6. List the names of the employees living in city Bombay and having salary >1500

Sample Code:

```
SELECT e.ename FROM URK24CS1021_employee e, URK24CS1021_emp_company c
WHERE e.ename = c.ename AND e.city = 'Bombay' AND c.salary > 1500;
```

Sample Output:

```
psql — labdb

labdb=# SELECT e.ename FROM URK24CS1021_employee e, URK24CS1021_emp_company c
labdb(# WHERE e.ename = c.ename AND e.city = 'Bombay' AND c.salary > 1500;
  ename
---------
 Shankar
 Sunil
(2 rows)

labdb=#
```

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7. Display the names of the employees living in the city Chennai and working with the company CMC.

Sample Code:

```
SELECT e.ename FROM URK24CS1021_employee e, URK24CS1021_emp_company c
WHERE e.ename = c.ename AND e.city = 'Chennai' AND c.cname = 'CMC';
```

Sample Output:

```
psql — labdb

labdb=# SELECT e.ename FROM URK24CS1021_employee e, URK24CS1021_emp_company c
labdb(# WHERE e.ename = c.ename AND e.city = 'Chennai' AND c.cname = 'CMC';
 ename
-------
 Jaya
(1 row)

labdb=#
```

8. Display the names of the employees living in Nagpur, working in Acc, and having shift A.

Sample Code:

```
SELECT ename FROM URK24CS1021_employee WHERE city = 'Nagpur'
INTERSECT
SELECT ename FROM URK24CS1021_emp_company WHERE cname = 'Acc'
INTERSECT
SELECT ename FROM URK24CS1021_emp_shift WHERE shift = 'A';
```

Sample Output:

```
psql — labdb

labdb=# SELECT ename FROM URK24CS1021_employee WHERE city = 'Nagpur'
labdb(# INTERSECT
labdb(# SELECT ename FROM URK24CS1021_emp_company WHERE cname = 'Acc'
labdb(# INTERSECT
labdb(# SELECT ename FROM URK24CS1021_emp_shift WHERE shift = 'A';
 ename
-------
 Anil
(1 row)

labdb=#
```

9. Display the names of the employees living in Nagpur, working in company Acc and not working in shift B.

Sample Code:

```
SELECT ename FROM URK24CS1021_employee WHERE city = 'Nagpur'
INTERSECT
```

```
SELECT ename FROM URK24CS1021_emp_company WHERE cname = 'Acc'
EXCEPT
SELECT ename FROM URK24CS1021_emp_shift WHERE shift = 'B';
```

Sample Output:

```
psql — labdb

labdb=# SELECT ename FROM URK24CS1021_employee WHERE city = 'Nagpur'
labdb(# INTERSECT
labdb(# SELECT ename FROM URK24CS1021_emp_company WHERE cname = 'Acc'
labdb(# EXCEPT
labdb(# SELECT ename FROM URK24CS1021_emp_shift WHERE shift = 'B';
 ename
-------
 Anil
(1 row)

labdb=#
```

10. Display the names of employees who are living in Nagpur or Bombay.

Sample Code:

```
SELECT ename FROM URK24CS1021_employee WHERE city = 'Nagpur'
UNION
SELECT ename FROM URK24CS1021_employee WHERE city = 'Bombay';
```

Sample Output:

```
psql — labdb

labdb=# SELECT ename FROM URK24CS1021_employee WHERE city = 'Nagpur'
labdb(# UNION
labdb(# SELECT ename FROM URK24CS1021_employee WHERE city = 'Bombay';
  ename
---------
 Ajay
 Anil
 Shankar
 Sunil
(4 rows)

labdb=#
```

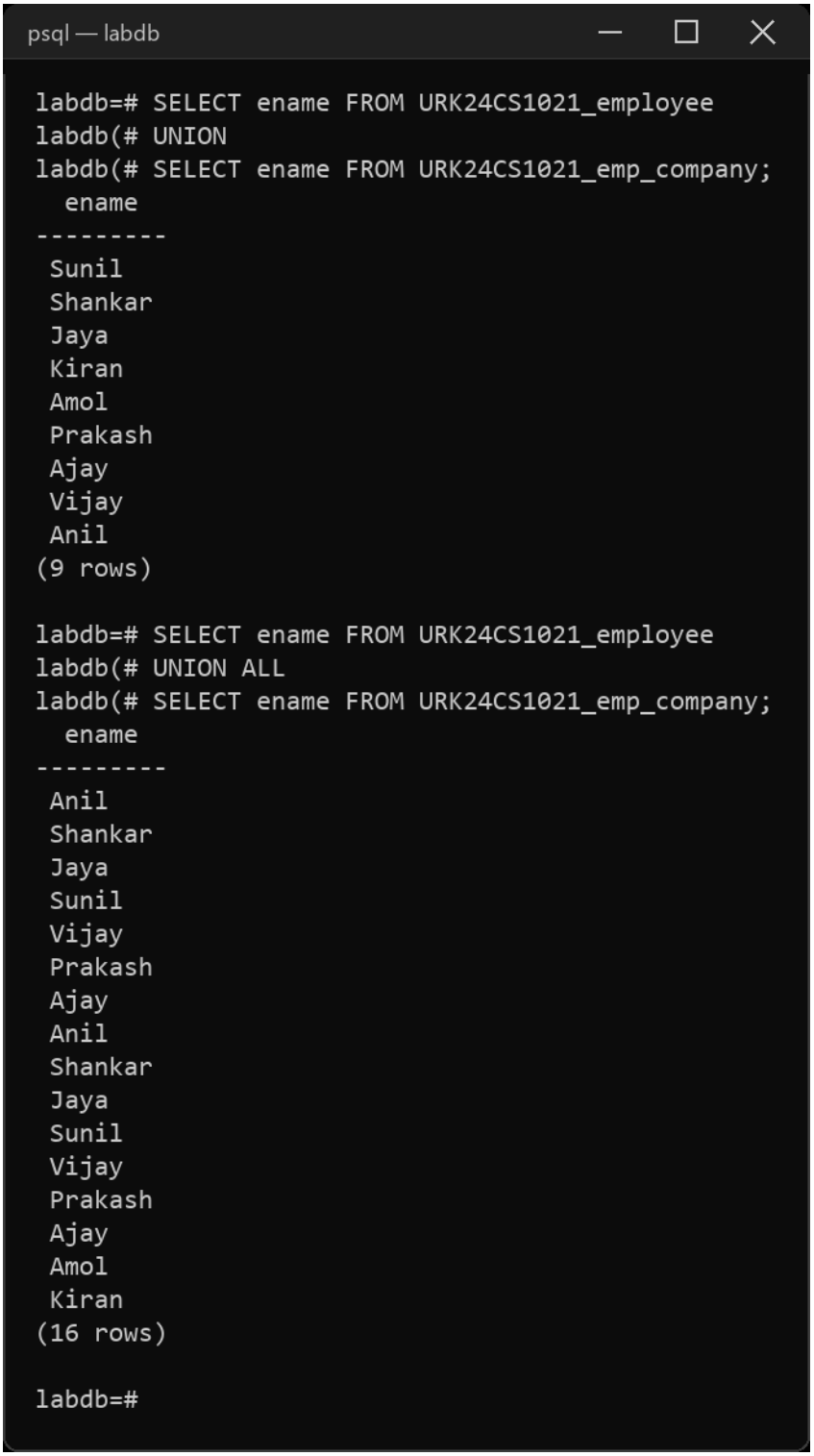
11. Display the employee names using union and union all operator.

Sample Code:

```
SELECT ename FROM URK24CS1021_employee
UNION
SELECT ename FROM URK24CS1021_emp_company;
SELECT ename FROM URK24CS1021_employee
UNION ALL
SELECT ename FROM URK24CS1021_emp_company;
```

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Sample Output:



12. Display the employee's name whose joining date is 18-DEC-98 and whose working shift is 'E'.

Sample Code:

```
SELECT ec.ename FROM URK24CS1021_emp_company ec, URK24CS1021_emp_shift s
WHERE ec.ename = s.ename AND ec.jdate = '1998-12-18' AND s.shift = 'E';
```

Sample Output:

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```
psql — labdb

labdb=# SELECT ec.ename FROM URK24CS1021_emp_company ec, URK24CS1021_emp_shift s
labdb(# WHERE ec.ename = s.ename AND ec.jdate = '1998-12-18' AND s.shift = 'E';
 ename
-------
 Kiran
(1 row)

labdb=#
```

13. Use in operator to display the names of the employees living in the same city where Sunil is residing.

Sample Code:

```
SELECT ename FROM URK24CS1021_employee
WHERE city IN (SELECT city FROM URK24CS1021_employee WHERE ename = 'Sunil');
```

Sample Output:

```
psql — labdb

labdb=# SELECT ename FROM URK24CS1021_employee
labdb(# WHERE city IN (SELECT city FROM URK24CS1021_employee WHERE ename = 'Sunil');
  ename
---------
 Shankar
 Sunil
(2 rows)

labdb=#
```

14. Use in operator to display the employees living in the city Bombay and having their company located in the city Delhi.

Sample Code:

```
SELECT ename FROM URK24CS1021_employee
WHERE city = 'Bombay' AND ename IN
    (SELECT ename FROM URK24CS1021_emp_company WHERE cname IN
        (SELECT cname FROM URK24CS1021_company WHERE city = 'Delhi'));
```

Sample Output:

```
psql — labdb

labdb=# SELECT ename FROM URK24CS1021_employee
labdb(# WHERE city = 'Bombay' AND ename IN
labdb(#     (SELECT ename FROM URK24CS1021_emp_company WHERE cname IN
labdb(#         (SELECT cname FROM URK24CS1021_company WHERE city = 'Delhi'));
 ename
-------
(0 rows)

labdb=#
```

15. Display the names of the employees having same city as Ajay [Hint: Use IN operator]

Sample Code:

```
SELECT ename FROM URK24CS1021_employee
WHERE city IN (SELECT city FROM URK24CS1021_employee WHERE ename = 'Ajay');
```

Sample Output:

```
psql — labdb

labdb=# SELECT ename FROM URK24CS1021_employee
labdb(# WHERE city IN (SELECT city FROM URK24CS1021_employee WHERE ename = 'Ajay');
 ename
-------
 Anil
 Ajay
(2 rows)

labdb=#
```

RESULT

The SQL queries were executed and the desired output was obtained.

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